Supervised Machine Learning - Classification (2-0-2-2-3)

Course Objectives

- Offers an in-depth overview of Machine learning models including ensemble models using real world data
- Develop algorithms using supervised learning methods
- Learn Python libraries for classification and ensemble models of machine learning

Course Outcome

The students by the end of the course will be able to:

- Explore the data using various python libraries.
- Understand Supervised classification including ensemble models and its applications.
- Build classification models and analyze model metrics/performance.

Pre-requisite Courses : Python for Data Science, Analytical Methods in Engineering Systems,

Course Content

1. **Supervised Learning – Classification – Logistic Regression (8 hours)** : Introduction to classification models, Binomial Logistic Regression, Assumptions of Logistic Regression, Significance of Coefficients, Model Evaluation Metrics, Model Performance Measures, Working with Imbalanced Data

   (Hands-on: Build a Binomial Logistic Regression model on an imbalanced dataset (like Credit Card Fraud Detection) and evaluate using Precision, Recall, F1-Score, and ROC Curve.)

2. **Naive Bayes, K - Nearest Neighbours (8 hours):** Naive Bayes, Fitting the K-NN algorithm to the Training set, Test accuracy of the result, Visualizing the result, Proximity Measures,

   (Hands-on: Implement Naive Bayes and KNN classification models on a text or numeric dataset (e.g., Iris / SMS Spam Detection) and visualize decision boundaries for KNN.)

3. **Decision trees (10 hours):**: Understanding Terminologies, Decision Trees for Classification, The measure of Purity of Node, Construction of Decision Tree, Decision Tree Algorithms, Model Evaluation, Model Performance Measures , Overfitting in a Decision Tree

   (Hands-on: Construct a Decision Tree classifier on a labeled dataset (e.g., Titanic Survival Prediction))

4. **Ensemble Learning (10 hours):** : Random Forest Classifier, Feature Importance, Bagging, Boosting Algorithms : Ada Boost, Gradient Boosting, AdaBoost Vs Gradient Boosting, XGBoost

(Hands-on:Apply Random Forest, AdaBoost, Gradient Boosting, and XGBoost on a classification dataset (like Bank Marketing / Telco Churn) and compare their performance.)

Reference books

1. Müller, A. C., Guido, S., & O'Reilly Media. (2018). Introduction to machine learning with Python: A guide for data scientists. Sebastopol: O'Reilly Media
2. Brunton, S. L., & Kutz, J. N. (2019). Data-driven science and engineering: Machine learning, dynamical systems, and control.
3. Shan, C., Wang, H. H., Chen, W., Song, M., & Klamka, J. (2015). The data science handbook: Advice and insights from 25 amazing data scientists.